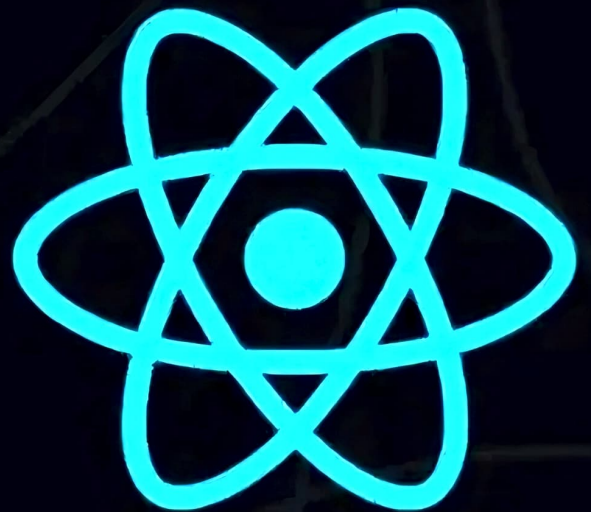


React vs JavaScript

What makes React Different
from JS ?





React is a **JavaScript library**, so it's not entirely different from JavaScript but rather an **extension** of it.

Lets see some key differences and how React extends JavaScript:

Component-Based Architecture

JavaScript :

JavaScript itself does not have a built-in way to structure UI components.

React :

Allows developers to build components that manage their **own state** and compose them to make **complex UIs**.

JSX

JavaScript :

Typically, **HTML** and **JavaScript** are written separately. **HTML** is used for structure, and **JavaScript** for behavior.

React :

uses **JSX**, a syntax extension that allows **HTML** to be written within **JavaScript**. This enables a more seamless integration of the **UI** structure with the logic.

Virtual DOM

JavaScript :

Manipulating the **DOM** directly can be slow and inefficient, especially with **large applications**.

React :

Uses a **virtual DOM** to optimize updates. It updates the virtual DOM first, **calculates the most efficient way** to apply changes to the real DOM, and then updates the real DOM, improving performance

State Management

JavaScript :

Managing state in a JavaScript application can become **complex** as the application grows.

React :

Provides **built-in mechanisms** for managing component state and propagating state changes, which helps in building interactive UIs.

Reusable Components

JavaScript :

Code can be modularized, but there is no structure for reusable components.

React :

Promotes the creation of reusable components, which can help in reducing redundancy and improving code maintainability.

Unidirectional Data Flow

JavaScript :

Data can flow in multiple directions, which can lead to complex and hard-to-maintain code.

React :

Controls a unidirectional data flow, making it easier to understand how data changes over time and how the application responds to those changes.